Lecture 22

# Type C tensor constructions and calculations

**Objectives.** *Lecture 21 found the fundamental symplectic modules inside exterior powers. We now find every irreducible $\mathfrak{sp}_{2n}$-module inside a tensor power of the defining module. Then we decompose small tensor products for $\mathfrak{sp}_4$.*

By the end of this lecture you will be able to:

- Prove $\operatorname{Sym}^k V \cong L(k\omega_1)$ by a highest vector and a dimension count.
- Define harmonic tensors and state the symplectic Weyl construction with its length hypothesis.
- Prove that $L(\lambda)$ occurs once in a tensor product of fundamental modules, and that the other summands have lower highest weights.
- Decompose small $\mathfrak{sp}_4$ tensor products and identify multiplicity spaces.

## 22.1 Symplectic data and highest vectors

Let $n \geq 1$ and $V = \mathbb{C}^{2n}$. All modules are finite-dimensional over $\mathbb{C}$. Tensor degrees are nonnegative integers. Fix a nondegenerate skew form $\beta$ on $V$. Choose an ordered basis

$$(e_1, \dots, e_n, f_1, \dots, f_n)$$

of $V$ such that

$$\beta(e_i, f_j) = \delta_{ij}, \qquad \beta(e_i, e_j) = \beta(f_i, f_j) = 0.$$

We write $\beta$ for the form, so that $\omega_k$ always denotes a fundamental weight. The *symplectic Lie algebra* is

$$\mathfrak{g} = \mathfrak{sp}_{2n} = \{X \in \mathfrak{gl}(V) : \beta(Xu, v) + \beta(u, Xv) = 0 \text{ for all } u, v \in V\}.$$

The constructions below hold for every $n \geq 1$. The coincidences $\mathfrak{sp}_2 = \mathfrak{sl}_2$ and $C_2 = B_2$ are recorded in Lecture 10.

The diagonal elements

$$H = \operatorname{diag}(t_1, \dots, t_n, -t_1, \dots, -t_n)$$

of $\mathfrak{g}$ form a Cartan subalgebra $\mathfrak{h}$. Put $\varepsilon_i(H) = t_i$. Then $e_i$ has weight $\varepsilon_i$ and $f_i$ has weight $-\varepsilon_i$. The data of Lecture 21 are

$$\begin{aligned} R_+ &= \{\varepsilon_i - \varepsilon_j,\ \varepsilon_i + \varepsilon_j\ (i < j),\ 2\varepsilon_i\}, \\ \alpha_i &= \varepsilon_i - \varepsilon_{i+1}\ (i < n), \qquad \alpha_n = 2\varepsilon_n, \\ \omega_k &= \varepsilon_1 + \cdots + \varepsilon_k, \qquad \rho = \textstyle\sum_{i=1}^{n} (n - i + 1)\varepsilon_i. \end{aligned}$$

For $\lambda = \sum_i \lambda_i \varepsilon_i$ the simple coroots give

$$\langle \lambda, \alpha_i^\vee \rangle = \lambda_i - \lambda_{i+1} \quad (i < n), \qquad \langle \lambda, \alpha_n^\vee \rangle = \lambda_n.$$

So $\lambda$ is dominant integral exactly when the $\lambda_i$ are integers with $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. Then

$$\lambda = \sum_{k=1}^{n} (\lambda_k - \lambda_{k+1}) \omega_k, \qquad \lambda_{n+1} = 0.$$

Thus $P_+$ corresponds to partitions with at most $n$ parts. Write $|\lambda| = \lambda_1 + \cdots + \lambda_n$, and write $(a_1, \ldots, a_n)$ for $\sum a_i \varepsilon_i$.

Write $\mu \leq \lambda$ when $\lambda - \mu$ lies in $Q_+$, the set of nonnegative integer combinations of positive roots. The positive coroots pair with $\ell = \sum_i \ell_i \varepsilon_i$ as

$$\langle \ell, (\varepsilon_i \pm \varepsilon_j)^\vee \rangle = \ell_i \pm \ell_j \quad (i < j), \qquad \langle \ell, (2\varepsilon_i)^\vee \rangle = \ell_i.$$

In the dimension formula of Lecture 17, multiply the factors for $\varepsilon_i - \varepsilon_j$ and $\varepsilon_i + \varepsilon_j$ into one factor. The formula then reads

$$\dim L(\lambda) = \prod_{i<j} \frac{\ell_i^2 - \ell_j^2}{\rho_i^2 - \rho_j^2} \prod_{i=1}^{n} \frac{\ell_i}{\rho_i}, \qquad \ell = \lambda + \rho, \quad \rho_i = n - i + 1. \tag{22.1}$$

**Lemma 22.1** (A highest vector generates a simple module). *Let $U$ be a finite-dimensional $\mathfrak{g}$-module. Let $v \in U$ be a nonzero vector of weight $\lambda$ with $\mathfrak{n}_+ v = 0$. Then the submodule $W$ generated by $v$ is isomorphic to $L(\lambda)$.*

*Proof.* By the PBW theorem (Lecture 11),

$$U(\mathfrak{g}) = U(\mathfrak{n}_-)\, U(\mathfrak{h})\, U(\mathfrak{n}_+).$$

The last two factors act on $v$ by scalars, since $\mathfrak{n}_+ v = 0$ and $v$ is a weight vector. So $W = U(\mathfrak{n}_-)v$. Hence every weight of $W$ is $\leq \lambda$, and $W_\lambda = \mathbb{C}v$.

The module $W$ is a finite-dimensional weight module generated by $v$. The cyclic-module lemma of Lecture 13 makes it simple. Its highest weight is $\lambda$, so $W \cong L(\lambda)$. ∎

## 22.2 Symmetric powers

**Theorem 22.2** (Symmetric powers). *For every $k \geq 0$, $\operatorname{Sym}^k V \cong L(k\omega_1)$, with highest vector $e_1^k$. In particular*

$$\dim L(k\omega_1) = \binom{2n + k - 1}{k}.$$

*Proof.* A monomial of degree $k$ in $\operatorname{Sym}^k V$ has weight

$$\operatorname{wt}\Big( \prod_{i=1}^{n} e_i^{a_i} f_i^{b_i} \Big) = \sum_{i=1}^{n} (a_i - b_i) \varepsilon_i, \qquad \sum_{i=1}^{n} (a_i + b_i) = k.$$

Its $\varepsilon_1$-coefficient satisfies

$$a_1 - b_1 \leq a_1 + b_1 \leq k,$$

with equality throughout only for $e_1^k$.

Let $\alpha \in R_+$ and $x \in \mathfrak{g}_\alpha$. Then $x \cdot e_1^k$ has weight $k\varepsilon_1 + \alpha$. Every positive root has $\varepsilon_1$-coefficient 0, 1 or 2. If it is 1 or 2, the coefficient of $k\varepsilon_1 + \alpha$ exceeds $k$, so this is not a weight. If it is 0, a weight with coefficient $k$ must equal $k\varepsilon_1$, which forces $\alpha = 0$. In both cases $x \cdot e_1^k = 0$.

Lemma 22.1 gives a submodule of $\operatorname{Sym}^k V$ isomorphic to $L(k\omega_1)$. It remains to compare dimensions.

Here $\ell = (k+n, n-1, \dots, 1)$, so $\ell_j = \rho_j$ for $j \geq 2$. In (22.1) only factors involving the index 1 survive. As $j$ runs from 2 to $n$, $m = \rho_j$ takes the values $n-1, \dots, 1$:

$$\dim L(k\omega_1) = \frac{k+n}{n} \prod_{m=1}^{n-1} \frac{(k+n-m)(k+n+m)}{(n-m)(n+m)}.$$

The four products are

$$\prod_{m=1}^{n-1}(k+n-m) = \frac{(k+n-1)!}{k!}, \quad \prod_{m=1}^{n-1}(k+n+m) = \frac{(k+2n-1)!}{(k+n)!},$$
$$\prod_{m=1}^{n-1}(n-m) = (n-1)!, \qquad \prod_{m=1}^{n-1}(n+m) = \frac{(2n-1)!}{n!}.$$

Substitute them. The factor $(k+n)\,(k+n-1)!$ cancels $(k+n)!$, and $n!/(n\,(n-1)!) = 1$:

$$\begin{aligned} \dim L(k\omega_1) &= \frac{k+n}{n} \cdot \frac{(k+n-1)!}{k!\,(n-1)!} \cdot \frac{(k+2n-1)!\,n!}{(k+n)!\,(2n-1)!} \\ &= \frac{(k+2n-1)!}{k!\,(2n-1)!} = \dim \operatorname{Sym}^k V. \end{aligned}$$

The last equality counts monomials of degree $k$ in $2n$ variables. The submodule is therefore all of $\operatorname{Sym}^k V$. ∎

For $k = 2$ the dimension is $n(2n+1) = \dim \mathfrak{g}$, and $2\varepsilon_1$ is the highest root. Exercise Sheet 22 builds the isomorphism $\operatorname{Sym}^2 V \cong \mathfrak{g}$ explicitly.

## 22.3 Harmonic tensors and Weyl's construction

For $1 \leq i < j \leq d$, the *pair contraction* is the linear map $C_{ij} \colon V^{\otimes d} \to V^{\otimes(d-2)}$ given on pure tensors by

$$C_{ij}(v_1 \otimes \cdots \otimes v_d) = \beta(v_i, v_j)\, v_1 \otimes \dots \widehat{v_i} \dots \widehat{v_j} \cdots \otimes v_d.$$

**Definition 22.3** (Harmonic tensors). A tensor $t \in V^{\otimes d}$ is *harmonic* if $C_{ij}(t) = 0$ for all $i < j$. Write

$$V_0^{\otimes d} = \bigcap_{i<j} \ker C_{ij}.$$

For $d \leq 1$ there are no contractions, and $V_0^{\otimes d} = V^{\otimes d}$.

**Proposition 22.4** (Stability of harmonic tensors). *The space $V_0^{\otimes d}$ is a $\mathfrak{g}$-submodule of $V^{\otimes d}$, stable under the permutation action of $S_d$. Every symmetric tensor is harmonic.*

*Proof.* An element $X \in \mathfrak{g}$ acts on a pure tensor by

$$X(v_1 \otimes \cdots \otimes v_d) = \sum_{r=1}^{d} v_1 \otimes \cdots \otimes Xv_r \otimes \cdots \otimes v_d.$$

Apply $C_{ij}$ to this sum. The terms with $r = i$ and $r = j$ contribute the tensor of the other $d-2$ factors times a scalar. By the definition of $\mathfrak{g}$, this scalar is

$$\beta(Xv_i, v_j) + \beta(v_i, Xv_j) = 0.$$

The terms with $r \neq i, j$ give

$$\sum_{r \neq i,j} C_{ij}(v_1 \otimes \cdots \otimes Xv_r \otimes \cdots \otimes v_d) = X\, C_{ij}(v_1 \otimes \cdots \otimes v_d).$$

Thus $C_{ij}$ is a $\mathfrak{g}$-map, and its kernel is a submodule.

Let $\sigma \in S_d$ act by

$$\sigma(v_1 \otimes \cdots \otimes v_d) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(d)}.$$

Positions $i$ and $j$ of $\sigma t$ carry the factors at positions $k = \sigma^{-1}(i)$ and $l = \sigma^{-1}(j)$ of $t$. Hence $C_{ij}(\sigma t) = \pm\sigma' C_{\{k,l\}}(t)$. Here $C_{\{k,l\}}$ contracts positions $k$ and $l$, taken in increasing order, and $\sigma'$ permutes the other $d-2$ positions. So $\sigma$ preserves $V_0^{\otimes d}$.

For the transposition $\tau = (i\,j)$, skewness of $\beta$ gives

$$C_{ij}(\tau t) = -C_{ij}(t) \qquad \text{for all } t \in V^{\otimes d}.$$

If $t$ is symmetric, then $\tau t = t$. Hence

$$C_{ij}(t) = C_{ij}(\tau t) = -C_{ij}(t),$$

so $C_{ij}(t) = 0$. ∎

Let $\lambda$ be a partition of $d$. Let $c_\lambda \in \mathbb{C}[S_d]$ be the Young symmetriser of Lecture 18. Use its image model $\mathbb{S}_\lambda V = c_\lambda V^{\otimes d}$.

**Definition 22.5** (Harmonic Schur module)**.** The *harmonic Schur module* is

$$\mathbb{S}_{\langle\lambda\rangle} V = V_0^{\otimes d} \cap \mathbb{S}_\lambda V.$$

The space $\mathbb{S}_{\langle\lambda\rangle} V$ is a $\mathfrak{g}$-submodule, since $c_\lambda$ commutes with $\mathfrak{g}$ and $V_0^{\otimes d}$ is a submodule. The identity

$$c_\lambda^2 = k_\lambda c_\lambda, \qquad k_\lambda \neq 0,$$

cited in Lecture 18, gives

$$\mathbb{S}_{\langle\lambda\rangle} V = c_\lambda V_0^{\otimes d}.$$

Indeed, if $t = c_\lambda s$ is harmonic, then

$$t = k_\lambda^{-1} c_\lambda^2 s = c_\lambda(k_\lambda^{-1} t) \in c_\lambda V_0^{\otimes d}.$$

Conversely, $c_\lambda V_0^{\otimes d}$ lies in $\mathbb{S}_\lambda V$. It also lies in $V_0^{\otimes d}$, since this space is stable under $S_d$ by Proposition 22.4.

**Theorem 22.6** (Symplectic Weyl construction)**.** *Let $V = \mathbb{C}^{2n}$ carry a nondegenerate skew form, and let $\lambda$ be a partition of $d \geq 0$.*

*(i) If $\lambda$ has at most $n$ parts, then $\mathbb{S}_{\langle\lambda\rangle} V \cong L(\lambda_1\varepsilon_1 + \cdots + \lambda_n\varepsilon_n)$.*

*(ii) If $\lambda$ has more than $n$ parts, then $\mathbb{S}_{\langle\lambda\rangle} V = 0$.*

· *C: Fulton–Harris, §17.3, Theorem 17.11; Goodman–Wallach, §10.2.3, Corollary 10.2.8.*

The sources state the result for the group $\mathrm{Sp}_{2n}(\mathbb{C})$. That group is connected, so its invariant subspaces in a tensor space coincide with the $\mathfrak{sp}_{2n}$-invariant ones (Fulton–Harris, Exercise 8.17). We use only this Lie algebra form.

Three remarks relate the theorem to earlier results.

- For $\lambda = (k)$, Proposition 22.4 gives $\mathbb{S}_{\langle k\rangle}V = \mathrm{Sym}^k V$. Theorem 22.2 proved this case independently.
- For $\lambda = (1^k)$, all contractions of an alternating tensor agree up to sign and permutation. So $\mathbb{S}_{\langle 1^k\rangle}V = \Lambda^k_0 V$, matching Lecture 21 for $k \le n$. For $k > n$ the partition $(1^k)$ has more than $n$ parts, so the theorem gives $\Lambda^k_0 V = 0$.
- The theorem does not declare kernels irreducible. For $n \ge 2$, already $V_0^{\otimes 2} = \mathrm{Sym}^2 V \oplus \Lambda^2_0 V$ has two nonzero summands. For $n = 2$, the kernel of $C_{12}$ on $V^{\otimes 3}$ has dimension $64 - 4 = 60$ and contains $\mathrm{Sym}^3 V$, of dimension 20.

## 22.4 Generation by fundamental modules

**Theorem 22.7** (Fundamental tensor generation). *Let $\lambda = \sum_k a_k\omega_k \in P_+$ and*

$$T = L(\omega_1)^{\otimes a_1} \otimes \cdots \otimes L(\omega_n)^{\otimes a_n},$$

*with $T = \mathbb{C}$ when $\lambda = 0$. Then $L(\lambda)$ is a summand of $T$ of multiplicity one. Every other simple summand has highest weight $\mu < \lambda$. Moreover $T$ embeds in $V^{\otimes|\lambda|}$.*

*Proof.* For $\lambda = 0$, the claim is $L(0) = \mathbb{C}$. Otherwise list the factors of $T$ as $L(\nu_1), \dots, L(\nu_N)$, with highest vectors $v_1, \dots, v_N$. The weights of $T$ are the sums $\mu_1 + \cdots + \mu_N$ with $\mu_r$ a weight of $L(\nu_r)$. Every weight of $L(\nu_r)$ is $\le \nu_r$, with one-dimensional $\nu_r$-space (Lecture 12). Note that $\sum_r \nu_r = \sum_k a_k\omega_k = \lambda$. Suppose $\mu_r \le \nu_r$ for each $r$ and $\sum_r \mu_r = \lambda$. Then

$$\sum_{r=1}^{N} (\nu_r - \mu_r) = \lambda - \lambda = 0, \qquad \nu_r - \mu_r \in Q_+.$$

A nonzero element of $Q_+$ has nonnegative simple-root coordinates, not all zero. So a vanishing sum forces $\mu_r = \nu_r$ for every $r$. Hence

$$T_\lambda = \mathbb{C}\, v, \qquad v = v_1 \otimes \cdots \otimes v_N,$$

and every weight of $T$ is $\le \lambda$.

Each $x \in \mathfrak{n}_+$ acts on $v$ as a sum of terms with $x$ applied to one $v_r$. Every term vanishes, so $\mathfrak{n}_+ v = 0$. Lemma 22.1 gives a submodule $U(\mathfrak{g})v \cong L(\lambda)$.

By Weyl's theorem on complete reducibility (Lecture 5), $T$ is a direct sum of simple modules. Each copy of $L(\lambda)$ in such a decomposition contributes one dimension to $T_\lambda$. So the multiplicity is one. The highest weight of any other summand is a weight of $T$, hence $\le \lambda$ and $\ne \lambda$.

For the embedding, $L(\omega_k) \cong \Lambda^k_0 V \subseteq \Lambda^k V$ by Lecture 21, and $\Lambda^k V$ is the submodule of alternating tensors in $V^{\otimes k}$. Tensoring these inclusions embeds $T$ in $V^{\otimes d}$ with

$$d = \sum_k k\, a_k = \sum_k k(\lambda_k - \lambda_{k+1}) = \sum_i \lambda_i = |\lambda|. \qquad \blacksquare$$

Thus every simple $\mathfrak{sp}_{2n}$-module $L(\lambda)$ lies in $V^{\otimes|\lambda|}$, in the degree predicted by Theorem 22.6. The proof constructs the highest vector. It does not decompose the lower summands.

## 22.5 Calculations for $\mathfrak{sp}_4$

Now $n = 2$ and $\rho = (2, 1)$. Write $L(a, b)$ for $L(a\varepsilon_1 + b\varepsilon_2)$. Formula (22.1) becomes

$$\dim L(a,b) = \frac{(\ell_1^2 - \ell_2^2)\,\ell_1\ell_2}{6}, \qquad (\ell_1, \ell_2) = (a+2, b+1).$$

The modules needed below are:

| $\lambda$ | $0$ | $\omega_1$ | $\omega_2$ | $2\omega_1$ | $\omega_1+\omega_2$ | $3\omega_1$ |
|---|---|---|---|---|---|---|
| $(a,b)$ | $(0,0)$ | $(1,0)$ | $(1,1)$ | $(2,0)$ | $(2,1)$ | $(3,0)$ |
| $\dim L(\lambda)$ | $1$ | $4$ | $5$ | $10$ | $16$ | $20$ |

For example, $(2, 1)$ gives $\ell = (4, 2)$ and $12 \cdot 8/6 = 16$.

**Example 22.8** (The square $V \otimes V$)**.** Theorem 22.2 and Lecture 21 give

$$V \otimes V = \operatorname{Sym}^2 V \oplus \Lambda^2 V \cong L(2\omega_1) \oplus L(\omega_2) \oplus \mathbb{C}, \qquad 16 = 10 + 5 + 1.$$

The characters confirm this. Group the weights into Weyl orbits. The table lists the multiplicity of each weight in an orbit:

| orbit | $V \otimes V$ | $L(2\omega_1)$ | $L(\omega_2)$ | $\mathbb{C}$ |
|---|---|---|---|---|
| $(\pm 2, 0), (0, \pm 2)$ | $1$ | $1$ | $0$ | $0$ |
| $(\pm 1, \pm 1)$ | $2$ | $1$ | $1$ | $0$ |
| $(0, 0)$ | $4$ | $2$ | $1$ | $1$ |

For instance, the zero weight of $\operatorname{Sym}^2 V$ is spanned by $e_1 f_1$ and $e_2 f_2$. Each row adds up.

The contraction $C_{12} \colon V \otimes V \to \mathbb{C}$ vanishes on $\operatorname{Sym}^2 V$ and on $\Lambda^2_0 V$. The invariant tensor

$$\Omega = \sum_{i=1}^{2} (e_i \otimes f_i - f_i \otimes e_i)$$

is the invariant bivector of Lecture 21, up to a nonzero scalar. It satisfies

$$C_{12}(\Omega) = \sum_{i=1}^{2} \big(\beta(e_i, f_i) - \beta(f_i, e_i)\big) = 2 + 2 = 4.$$

So $\Omega$ is a nonzero invariant tensor outside $\operatorname{Sym}^2 V \oplus \Lambda^2_0 V$, and it spans the trivial summand. For every $\mathfrak{sp}_{2n}$, the invariant tensor lies in the alternating part $\Lambda^2 V$.

**Example 22.9** (The product $V \otimes L(\omega_2)$)**.** Weights of $L(\omega_2)$ are $(\pm 1, \pm 1)$ and $(0, 0)$, each once. Add the weights $(\pm 1, 0), (0, \pm 1)$ of $V$. The weight $(2, 1)$ arises only as $(1, 0) + (1, 1)$. The weight $(1, 0)$ arises three times:

$$(1,0) + (0,0), \qquad (0,1) + (1,-1), \qquad (0,-1) + (1,1).$$

By Weyl invariance (Lecture 15), the 8 weights of the orbit of $(2, 1)$ have multiplicity 1. The 4 weights of the orbit of $(1, 0)$ have multiplicity 3. This accounts for $8 + 12 = 20$.

By Theorem 22.7, $L(\omega_1 + \omega_2)$ occurs once. Elements of $Q_+$ have the form

$$p\alpha_1 + q\alpha_2 = (p,\ 2q - p), \qquad p, q \in \mathbb{Z}_{\geq 0}.$$

The dominant weights of $V \otimes L(\omega_2)$ are $(2,1)$ and $(1,0)$, and

$$(2,1) - (1,0) = (1,1) = \alpha_1 + \alpha_2 \in Q_+.$$

Since $L(\omega_1 + \omega_2)$ is a summand, its dominant weights are among these two. If $m$ is the multiplicity of $(1,0)$ in $L(\omega_1 + \omega_2)$, then

$$16 = 8 \cdot 1 + 4m, \quad \text{so} \quad m = 2.$$

Subtracting characters gives

$$\begin{aligned} \operatorname{ch}(V \otimes L(\omega_2)) - \operatorname{ch} L(\omega_1 + \omega_2) &= \sum_{\mu=(\pm 1,0),(0,\pm 1)} (3-2)\, e^{\mu} \\ &= e^{(1,0)} + e^{(-1,0)} + e^{(0,1)} + e^{(0,-1)} = \operatorname{ch} V. \end{aligned}$$

The orbit of $(2,1)$ cancels completely. A finite-dimensional module is determined by its character (Lecture 17). Hence

$$V \otimes L(\omega_2) \cong L(\omega_1 + \omega_2) \oplus L(\omega_1), \qquad 20 = 16 + 4.$$

Every weight here has odd coordinate sum, so $L(\omega_2)$ and $\mathbb{C}$ cannot occur.

A contraction realises the small summand. Define the linear map $c\colon V \otimes \Lambda^2 V \to V$ by

$$c(u \otimes x \wedge y) = \beta(u,x)\, y - \beta(u,y)\, x.$$

The right-hand side is alternating in $x$ and $y$, so $c$ is well defined. Invariance of $\beta$ makes $c$ a $\mathfrak{g}$-map, as in Proposition 22.4. Since $\beta(e_1, e_2) = 0$, the vector $e_1 \wedge e_2$ lies in $\Lambda^2_0 V$, and

$$c(f_1 \otimes e_1 \wedge e_2) = \beta(f_1, e_1)\, e_2 - \beta(f_1, e_2)\, e_1 = -e_2.$$

The image of $c$ is a nonzero submodule of the simple module $V$. So $c$ maps $V \otimes \Lambda^2_0 V$ onto $V$, and its kernel has dimension

$$20 - 4 = 16.$$

The two summands of $L(\omega_1 + \omega_2) \oplus L(\omega_1)$ are nonisomorphic. So every submodule is a sum of some of them, and the only one of dimension 16 is $L(\omega_1 + \omega_2)$. Hence the kernel of $c$ on $V \otimes \Lambda^2_0 V$ is irreducible. This is a computed fact, not a general property of kernels.

**Example 22.10** (Multiplicity spaces in $V^{\otimes 3}$)**.** First decompose $V \otimes \operatorname{Sym}^2 V$, of dimension $4 \cdot 10 = 40$. Its dominant weights are $(3,0)$, $(2,1)$ and $(1,0)$. The top weight $(3,0)$ occurs once, from $e_1 \otimes e_1^2$. So $L(3\omega_1) \cong \operatorname{Sym}^3 V$ occurs once. The weight $(2,1)$ has the following bases:

$$\begin{aligned} V \otimes \operatorname{Sym}^2 V: &\quad e_1 \otimes e_1 e_2, \quad e_2 \otimes e_1^2 \quad \text{(dimension 2)}, \\ \operatorname{Sym}^3 V: &\quad e_1^2 e_2 \qquad\qquad\quad \text{(dimension 1)}. \end{aligned}$$

So $L(\omega_1 + \omega_2)$ occurs $2 - 1 = 1$ times. Since $(3,0) \not\leq (2,1)$, the module $L(\omega_1 + \omega_2)$ has no weight $(3,0)$. Hence the weights $(3,0)$ and $(2,1)$ are now exhausted. The remaining

$$40 - 20 - 16 = 4$$

dimensions have highest weight $(1,0)$ only, so they form one copy of $L(\omega_1)$.

With Examples 22.8 and 22.9,

$$\begin{aligned} V^{\otimes 3} &= V \otimes \operatorname{Sym}^2 V \;\oplus\; V \otimes \Lambda^2_0 V \;\oplus\; V \otimes \mathbb{C} \\ &\cong L(3\omega_1) \oplus L(\omega_1 + \omega_2)^{\oplus 2} \oplus L(\omega_1)^{\oplus 3}, \end{aligned}$$

with $64 = 20 + 32 + 12$. By Schur's lemma, the multiplicity of $L(\omega_1)$ is

$$\dim \mathrm{Hom}_{\mathfrak{g}}(V, V^{\otimes 3}) = 3.$$

Three explicit intertwiners $V \to V^{\otimes 3}$ insert $\Omega$ into two of the three positions:

$$\begin{aligned}
\iota_{12}(v) &= \Omega \otimes v, \\
\iota_{13}(v) &= \textstyle\sum_i (e_i \otimes v \otimes f_i - f_i \otimes v \otimes e_i), \\
\iota_{23}(v) &= v \otimes \Omega.
\end{aligned}$$

They are $\mathfrak{g}$-maps because $\Omega$ is invariant and permutations commute with $\mathfrak{g}$.

The three maps are linearly independent. Evaluate them at $e_1$. The tensors

$$e_2 \otimes f_2 \otimes e_1, \qquad e_2 \otimes e_1 \otimes f_2, \qquad e_1 \otimes e_2 \otimes f_2$$

occur only in $\iota_{12}(e_1)$, $\iota_{13}(e_1)$ and $\iota_{23}(e_1)$ respectively. So a vanishing combination of the three maps has zero coefficients.

Evaluation (Lecture 14) identifies the $L(\omega_1)$-isotypic component with

$$\mathrm{Hom}_{\mathfrak{g}}(V, V^{\otimes 3}) \otimes V \cong \mathbb{C}^3 \otimes V.$$

## Self-checks

(1) What is $\dim L(3\omega_1)$ for $\mathfrak{sp}_6$?
(2) For $\mathfrak{sp}_4$, what is $\mathbb{S}_{\langle 1,1,1\rangle} V$, and what is $\Lambda^3 V$?
(3) Why can $\mathbb{C}$ not occur in $V \otimes L(\omega_2)$ for $\mathfrak{sp}_4$?

*Answers.* (1) By the theorem on symmetric powers, it is $\dim \mathrm{Sym}^3 \mathbb{C}^6 = \binom{8}{3} = 56$. (2) The module $\mathbb{S}_{\langle 1,1,1\rangle} V$ is zero, since $(1,1,1)$ has three parts and $n = 2$. The weights of $\Lambda^3 V$ are $\pm\varepsilon_1, \pm\varepsilon_2$, each once. So $\Lambda^3 V \cong L(\omega_1)$, a copy containing no nonzero harmonic tensor. (3) All weights of $V \otimes L(\omega_2)$ have odd coordinate sum. The trivial module $\mathbb{C}$ has only the weight 0, of even sum.

## Summary

Symmetric powers of $V$ are irreducible: $\mathrm{Sym}^k V \cong L(k\omega_1)$. In general, the harmonic Schur module intersects $\mathbb{S}_\lambda V$ with the kernels of all pair contractions. The symplectic Weyl construction identifies it with $L(\lambda)$ when $\lambda$ has at most $n$ parts, and gives zero otherwise.

A tensor product of fundamental modules contains $L(\lambda)$ once, where $\lambda$ is the sum of their highest weights. The tensor product of highest vectors generates this summand.

For $\mathfrak{sp}_4$, characters and contractions decompose small products. A repeated summand has an explicit multiplicity space, such as $\mathrm{Hom}_{\mathfrak{g}}(V, V^{\otimes 3})$.

## Reading

William Fulton and Joe Harris, *Representation Theory: A First Course*, §17.3. Roe Goodman and Nolan R. Wallach, *Symmetry, Representations, and Invariants*, Chapter 10. Roger Carter, *Lie Algebras of Finite and Affine Type*, Chapter 13. Exercise Sheet 22 gives further calculations.